Exploring Mixtures and Their Separation

Important Questions with Answers

Class 9 Science | NCERT | 42 Questions

MCQ - True/False - Fill in the blanks - Short answer - Numericals - Long answer

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Tip: cover the green answer box, try the question first, then check. Compare with your own textbook wording.

A. Multiple Choice Questions

Q1 - Q4

Q1. Which of the following is a **heterogeneous** mixture?
(a) Salt solution (b) Air (c) Muddy water (d) Brass

Ans: (c) Muddy water. Its components (soil and water) can be seen separately and are not uniformly mixed.

Q2. Which of these does **not** show the Tyndall effect?
(a) Milk (b) Fog (c) Sugar solution (d) Starch solution

Ans: (c) Sugar solution. It is a true solution, so its particles are too small to scatter light.

Q3. The best method to separate cream from milk is:
(a) Filtration (b) Centrifugation (c) Sublimation (d) Evaporation

Ans: (b) Centrifugation. Fast spinning forces the denser particles to the bottom and the lighter cream collects on top.

Q4. Two miscible liquids with boiling points close to each other are separated by:
(a) Decantation (b) Sieving (c) Simple distillation (d) Fractional distillation

Ans: (d) Fractional distillation. The fractionating column gives repeated condensation and evaporation, which separates liquids with close boiling points.

A. Multiple Choice Questions (contd.)

Q5. 10 g of salt is dissolved in 90 g of water. The mass percentage of the solution is:
(a) 9% (b) 10% (c) 11.1% (d) 90%

Ans: (b) 10%. Mass of solution = 10 + 90 = 100 g, so mass % = (10 / 100) x 100 = 10%.

Q6. Which of the following is a pure substance?
(a) Air (b) Sea water (c) Distilled water (d) Milk

Ans: (c) Distilled water. It contains only one kind of particle (H_2O). The others are mixtures.

Q7. A mixture of ammonium chloride and sand can be separated by:
(a) Sublimation (b) Magnetic separation (c) Winnowing (d) Sedimentation

Ans: (a) Sublimation. Ammonium chloride changes directly from solid to gas on heating. Sand does not.

Q8. In a solution, the component present in the larger amount is called the:
(a) Solute (b) Solvent (c) Precipitate (d) Colloid

Ans: (b) Solvent. The component present in the smaller amount is the solute.

B. True or False

Q9 - Q13

Q9. Air is a homogeneous mixture of gases.

Ans: TRUE. Its composition is uniform throughout (in clean air).

Q10. The particles of a suspension settle down when the mixture is left undisturbed.

Ans: TRUE. The particles are large (over 100 nm) and heavy, so they settle.

Q11. A colloid can be separated into its components by ordinary filtration.

Ans: FALSE. Colloidal particles are small enough to pass through filter paper. Centrifugation is used instead.

Q12. Milk is a colloid.

Ans: TRUE. Tiny fat droplets are spread through water and they scatter light.

Q13. The solubility of gases in water increases when the temperature rises.

Ans: FALSE. It decreases. That is why warm soft drinks lose fizz faster.

B. True or False (contd.)

Q14. A mixture has a fixed composition.

Ans: FALSE. The components can be mixed in any proportion. A compound has a fixed composition.

Q15. The components of a mixture keep their own properties.

Ans: TRUE. No new substance forms, so each component keeps its identity.

Q16. Brass is a heterogeneous mixture.

Ans: FALSE. Brass is an alloy (copper and zinc) and is a homogeneous mixture.

Q17. Distillation separates liquids on the basis of the difference in their boiling points.

Ans: TRUE. The liquid with the lower boiling point vaporises first.

Q18. A saturated solution can still dissolve more solute at the same temperature.

Ans: FALSE. By definition it cannot dissolve any more solute at that temperature.

C. Fill in the Blanks

Q19. A solution in which water is the solvent is called an _____ solution.

Ans: Aqueous solution.

Q20. Separating lighter husk from heavier grains using wind is called _____.

Ans: Winnowing.

Q21. The scattering of light by colloidal particles is called the _____ effect.

Ans: Tyndall effect.

C. Fill in the Blanks (contd.)

Q22. In chromatography, different components move at different _____ on the paper, so they separate.

Ans: Speeds (rates). They differ in solubility in the solvent and in attraction to the paper.

Q23. The process in which a solid changes directly into vapour on heating is called _____.

Ans: Sublimation. Examples: ammonium chloride, camphor, naphthalene.

Q24. Iron filings can be separated from a mixture with sulphur powder by using a _____.

Ans: Magnet (magnetic separation), because iron is attracted to a magnet.

D. Short Answer Questions

Q25 - Q28

Q25. What is a mixture? Give one example.

Ans: A mixture is made of two or more substances (elements or compounds) physically mixed in any proportion, with no chemical bonding. Example: sugar and sand, or air.

Q26. Differentiate between homogeneous and heterogeneous mixtures.

Ans: Homogeneous: uniform composition throughout, one phase (salt solution, air, brass). Heterogeneous: non-uniform composition, parts are visibly different (sand and water, oil and water).

Q27. What is the Tyndall effect? Why does a true solution not show it?

Ans: It is the scattering of a light beam by particles so that the path of light becomes visible. Particles of a true solution are smaller than 1 nm, too small to scatter visible light.

Q28. Differentiate between a solute and a solvent.

Ans: The solute is the component present in the smaller amount, which dissolves. The solvent is the component present in the larger amount, which dissolves the solute. In sugar solution, sugar is the solute and water is the solvent.

D. Short Answer Questions (contd.)

Q29. Define solubility and a saturated solution.

Ans: Solubility is the maximum amount of solute that dissolves in a given amount of solvent at a given temperature. A saturated solution has dissolved that maximum amount and cannot take in more at that temperature.

Q30. What is decantation? When is it used?

Ans: Allowing a mixture to stand so the heavier insoluble solid settles (sedimentation), then carefully pouring off the clear liquid. Used for muddy water and for separating a liquid from settled solids.

Q31. Write the formulae for mass percentage and volume percentage of a solution.

Ans: Mass % = (mass of solute / mass of solution) x 100. Volume % = (volume of solute / volume of solution) x 100. Here, solution = solute + solvent.

Q32. Why is crystallisation preferred over simple evaporation to purify a solid?

Ans: Evaporation to dryness can decompose or char some solids (e.g. sugar) and leaves soluble impurities mixed in. Crystallisation gives pure, well-formed crystals while impurities stay in the liquid.

D. Short Answer Questions (contd.)

Q33. Differentiate between an element, a compound and a mixture.

Ans: Element: one kind of atom (iron, oxygen). Compound: two or more elements chemically combined in a fixed ratio (water). Mixture: substances physically mixed in any ratio, separable by physical methods.

Q34. What is an alloy? Give an example.

Ans: An alloy is a homogeneous mixture of a metal with other metals or non-metals. Example: brass (copper and zinc), steel (iron and carbon).

E. Numericals

Q35. A solution contains 20 g of salt dissolved in 180 g of water. Find the mass percentage of the solution.

Ans: Mass of solution = 20 + 180 = 200 g.
Mass % = (20 / 200) x 100 = **10%**.

Q36. 50 mL of alcohol is dissolved in 200 mL of water. Find the volume percentage of alcohol (assume volumes add up).

Ans: Volume of solution = 50 + 200 = 250 mL.
Volume % = (50 / 250) x 100 = **20%**.

Q37. How much sugar and water are needed to prepare 250 g of a 12% (by mass) sugar solution?

Ans: Sugar = 12% of 250 = (12 / 100) x 250 = **30 g**.
Water = 250 - 30 = **220 g**.

Q38. Describe how you would separate a mixture of sand, salt and iron filings. Name each technique used.

Answer

Step 1 - Magnetic separation: Bring a magnet near the dry mixture. Iron filings are attracted and stick to it. Sand and salt are not.

Step 2 - Dissolving: Add the remaining sand and salt to water and stir. Salt dissolves. Sand does not (difference in solubility).

Step 3 - Filtration: Pour the mixture through filter paper in a funnel. Sand remains on the paper as the *residue*. The salt solution passes through as the *filtrate*.

Step 4 - Evaporation: Heat the filtrate to evaporate the water. Salt is left in the dish. (Crystallisation gives purer salt.)

Result: Iron filings, sand and salt are all obtained separately using only physical methods.

Q39. Explain fractional distillation. Why is it needed to separate liquids whose boiling points are close?

Answer

Use: Separating a mixture of miscible liquids whose boiling points are close (difference less than about 25 K).

Set-up: A fractionating column, packed with glass beads or rings, is fitted between the distillation flask and the condenser.

Working: (1) Heat the mixture so that vapours rise into the column. (2) The beads give a large surface area. The vapours condense and re-evaporate many times. (3) The vapour of the liquid with the *lower boiling point* reaches the top first, passes into the condenser and is collected. (4) The liquid with the higher boiling point condenses and falls back into the flask.

Why needed: In simple distillation both liquids vaporise together, so the separation is poor. The column repeats the evaporation-condensation cycle and improves it.

Examples: Separating the gases of air from liquid air; separating the fractions of crude oil (petrol, diesel, kerosene).

Q40. Differentiate between a solution, a colloid and a suspension.

Property	Solution	Colloid	Suspension
Particle size	Less than 1 nm	1 nm to 100 nm	More than 100 nm
Appearance	Transparent, clear	Looks uniform but is heterogeneous (translucent)	Opaque or cloudy, heterogeneous
Stability	Stable, particles do not settle	Stable, particles do not settle	Unstable, particles settle on standing
Tyndall effect	Not shown	Shown	Shown
Filtration	Passes through filter paper	Passes through filter paper	Residue stays on filter paper
Examples	Salt in water, air, brass	Milk, fog, butter, smoke	Muddy water, chalk in water

Q41. Describe the process of crystallisation. How is it better than simple evaporation?

Answer

Use: To purify a solid that dissolves in a hot solvent, e.g. impure copper sulphate or alum.

Step 1: Dissolve the impure solid in the minimum amount of hot water to make a saturated solution.

Step 2: Filter the hot solution to remove insoluble impurities.

Step 3: Evaporate some water to concentrate the solution, then leave it to cool slowly without disturbing it.

Step 4: Crystals form. Filter them out, wash with a little cold water and dry between filter papers.

Better than evaporation because: (i) heating to dryness can decompose or char some solids (e.g. sugar); (ii) soluble impurities stay behind in the liquid (mother liquor), giving pure crystals; (iii) evaporation leaves all impurities mixed with the solid.

Q42. What is the principle of chromatography? Describe a simple paper chromatography experiment and give its uses.

Answer

Principle: Different components of a mixture dissolve in the solvent to different extents and are held on the paper with different strength. So they travel at different speeds and separate.

Procedure: (1) Draw a pencil line near the bottom of a filter paper strip and put a small spot of black ink on it. (2) Dip the lower edge of the strip in water, keeping the spot above the water level. (3) Water rises up the paper by capillary action and carries the dyes along at different speeds. (4) Remove the strip when the water is near the top and let it dry. Coloured spots appear at different heights.

Why pencil? Graphite does not dissolve in the solvent, but ink would run and mix.

Uses: Separating dyes in ink, pigments in plant leaves, colours in food items; checking the purity of a substance; detecting drugs or poisons in forensic work.

Quick Revision: Separation Methods

Method	Used to separate	Based on
Handpicking / Sieving	Large solids from small ones	Size of particles
Winnowing	Husk from grain	Weight (wind blows lighter husk away)
Magnetic separation	Iron from non-magnetic solids	Magnetic property
Sedimentation + Decantation	Insoluble heavy solid from liquid	Density, settling
Filtration	Insoluble solid from liquid	Particle size vs filter pores
Centrifugation	Cream from milk, blood cells	Density (spinning)
Evaporation	Dissolved solid from solvent	Volatility of solvent
Crystallisation	Pure crystals from solution	Solubility changes with temperature
Sublimation	Ammonium chloride, camphor from others	Solid changes directly to gas
Distillation	Liquid from dissolved solid, miscible liquids	Boiling point difference
Separating funnel	Immiscible liquids (oil and water)	Liquids do not mix, different density
Chromatography	Dyes, pigments in a mixture	Different speeds on the paper